

Visual Augmentation of Audio Semantics for Accessibility

Technical report

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Abstract

Subtitles carry speech; the ambient sound of a video – a siren behind the camera, a dog barking in the next room, glass breaking off screen – reaches a deaf viewer only if a captioner chose to write it down. This project asks whether that channel can be automated: given a video, detect its ambient sounds, decide for each whether its source is already visible on screen (this decision is the *gate*), and show a picture of the event beside the video only while a sound whose source is not visible is heard. The system is an inference pipeline over open-source foundation models with no training: BEATs, an audio tagging model, for detection, a vision-language model for the visibility decision and the depiction, and a text-to-image model for the picture. It is evaluated on a benchmark of 274 hand-labelled clips in four scenarios against a blind audio-to-image baseline and an audio-captioning baseline, with an automatic protocol in which a vision-language model describes the augmented video and a judge scores it against a reference. Under the human-grounded reference the gated system ties the blind baseline overall (3.17 vs 3.16 on a 0–4 scale, 100 clips), beats it where no picture is due (+0.26; the 95% confidence interval excludes zero) and loses where one is (−0.24); a perfect gate would score 3.25. Three findings about the evaluation itself are reported alongside. First, the reference specified in the proposal is written by the system’s own models and biases the comparison by half a point. Second, a reference built by independent models cannot decide whether a source is on screen better than chance. Third, the rubric is asymmetric – a withheld picture scores 0, a redundant one about 3 – so it penalises staying silent, which is the behaviour the gate exists to provide.

1 Introduction

Deaf and hard-of-hearing viewers receive speech through subtitles and, where available, sign language. Neither channel is built for the rest of the soundtrack. Subtitles for the deaf and hard-of-hearing (SDH) may name a sound effect in brackets, but style guides such as the DCMF Captioning Key [24] instruct captioners to omit any sound that can be inferred from the picture, and studies of caption use find non-speech information inconsistently rendered and often absent [19]. Sign-language interpretation is expressive but scarce and not automatable. Field studies of sound-awareness systems [22, 21] show that deaf users want to know about exactly the sounds that carry information – knocks, alarms, sirens, a baby crying, a dog barking – and that they read icons and pictures faster than text.

The information a hearing viewer gets from ambient sound is therefore lost in proportion to how far the sound’s source is from the frame. A siren over a visible ambulance is redundant; a siren over a crowd looking up is the scene. This project targets that second case. It differs from audio-to-image generation [14, 16], which turns sound into a picture regardless of the video, and from sound-source localisation [28, 30], which finds where in the frame a sound comes from without saying what should be done about it. The contribution is the combination: detect the ambient sound, decide from the video whether its source is already visible, and show a picture of the event only when it is not.

1.1 Problem statement

Given a video, produce the same video with a side panel that shows, for the duration of each ambient sound whose source is not visible on screen, a picture of that sound’s event – and shows nothing otherwise. Speech and music are excluded. No model is trained; the system is an inference pipeline over open-source foundation models, and human involvement is limited to the benchmark labels used for evaluation.

1.2 Research questions

1. Which types of audio information contribute most to scene understanding when presented visually?
2. What level of semantic granularity should be used to represent ambient sounds?
3. Can automatically generated visual augmentations improve accessibility beyond subtitles?
4. How should semantic audio augmentation systems be evaluated?
5. How does the proposed pipeline compare with representative audio-to-visual approaches?

1.3 Contributions

- Formulation of semantic audio augmentation as an accessibility task, with a precise definition of when a picture is due (sound heard, source not visible, not speech or music).
- A seven-stage inference pipeline of off-the-shelf models (§3).
- A benchmark of 274 human-labelled clips in four scenarios (§4).
- An automatic evaluation protocol, and the finding that its reference must be grounded in labels: one written by the system’s own models is circular, and one written by independent models cannot decide what is on screen (§5).
- An experimental comparison with two baselines, with paired statistics and an oracle bound (§6).

2 Related work

Audio event detection. Nearly all modern taggers are trained on AudioSet, a dataset of two million clips labelled with a hierarchy (ontology) of 527 sound classes [1]. PANNs [2] established strong CNN features (0.431 mean average precision, mAP); transformer taggers – AST [3] and BEATs [4], the latter pre-trained with acoustic tokenizers – lead the DCASE (Detection and Classification of Acoustic Scenes and Events) challenges (BEATs 0.486 mAP on the same labels). BEATs scores a whole clip at once (clip-level); detectors that score every frame locate a sound’s start (its *onset*) more sharply but miss more sounds, which §6 measures.

Audio captioning and audio language models. Captioning systems trained on AudioCaps [5] and Clotho [6] describe a soundscape in a sentence, and audio-language models such as Qwen2-Audio [9] and SALMONN [10] answer questions about it. Text is the natural competitor to a picture and is the second baseline here; an audio LM is used only inside the evaluation, to write a reference that never saw the system. CLAP [7] embeds audio and text in one space and is used to verify the audio LM’s claims.

Vision–language models. Qwen2.5-VL [12], LLaVA [13] and Idefics3 answer open questions about frames. They are the only component that can decide whether a sound’s source is visible without a fixed vocabulary of objects, and §3 shows that asking the model to *name* the source,

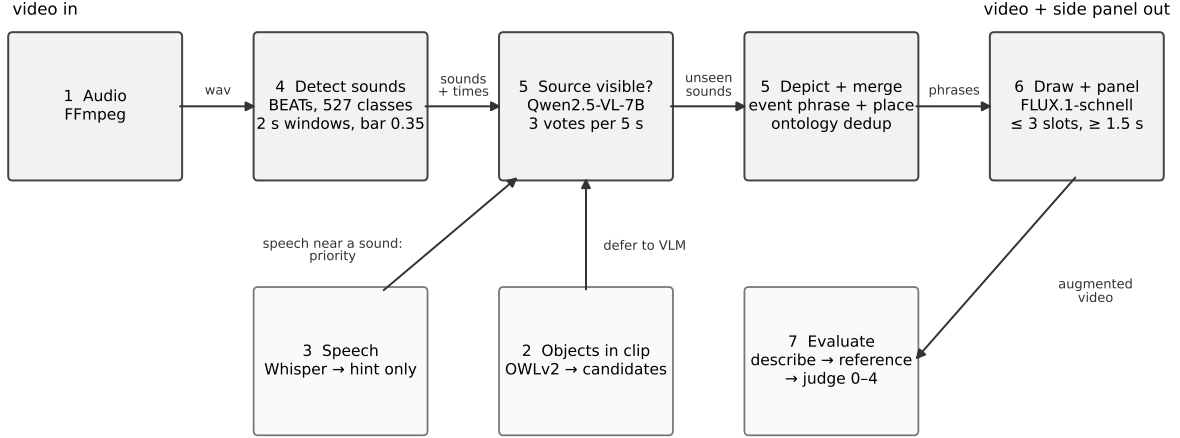


Figure 1: The seven stages and what flows between them. Stages 2 and 3 only feed the visibility decision; stage 7 is the evaluation of §5.

and asking each two-way question in both orders, is what makes its answer evidence rather than mere agreement.

Sound-source localisation. Cross-modal alignment methods [28, 29] point to where in an image a sound originates, and off-screen sound separation [30] estimates whether a source is in frame at all. They answer the gate’s question with a number rather than a judgement; they were not used here because they are trained on data where the source usually is in frame, and the clips this benchmark cares about are exactly those where it is not. The DCASE 2025 Task 3 dataset, which labels every event on- or off-screen, is used to check the VLM gate instead (§4).

Audio-to-image and audio-to-video. Sound2Scene [14] maps audio into a pretrained image generator’s latent space; SonicDiffusion [16] conditions a diffusion model on audio; TempoTokens [15] adapts a text-to-video model to audio. All are blind to the existing video and are the family this work is measured against. The picture itself is drawn by a text-conditioned model (FLUX.1 [18]; SDXL [17] in earlier runs), with the prompt chosen by the reasoning stage rather than from the audio alone.

Accessibility studies. Non-text sound visualisations – icons, directional indicators, animated effects – are learnable and preferred by deaf viewers in games and AR [23], and author-selected sound visuals are welcomed in media; caption research [26, 25, 27] documents the cost of divided attention, which is why the panel here keeps a fixed position and shows nothing when nothing is heard. What none of this work does is generate imagery for ambient sound in recorded media, selected automatically and gated by what is already visible.

3 Method

3.1 Pipeline

stage	model	role
1 audio extraction	FFmpeg	16 kHz mono wav
2 video understanding	OWLv2 (fast object detector)	candidate visible objects
3 speech recognition	Whisper (configurable)	transcript, used only as reference
4 audio event detection	BEATs iter3+ AS2M (527 AudioSet classes)	events with onsets
5 cross-modal reasoning	Qwen2.5-VL-7B	place, visibility, kind, color
6 visual augmentation	FLUX.1-schnell, 768 px	one picture per sound, 100px
7 evaluation	Qwen2.5-VL / Mistral-7B judge + independent reference	§5

3.2 Stage 4: detection and onset timing

Detector. Stage 4 tags audio with BEATs (iter3+, fine-tuned on AudioSet-2M; 0.486 mAP against PANNs CNN14’s 0.431 on the same 527 labels). BEATs scores a whole clip at once, so the pipeline scores a 2-second window every 0.25s, giving one score per time step. The start of the audio is padded by mirroring its first two seconds, so a sound at the very beginning can still be detected. Each window’s score is assigned to the time 0.5s before the window ends, so a detected start is never earlier than the true one. The swap from PANNs was made after a review of eight demo clips traced eleven of fourteen complaints to the detector’s labels (a siren as “truck horn”, a crying baby as “sheep”); on the same clips every one resolved. Speech and music classes are never shown. The display threshold (the *bar*: the score a sound must reach to be shown) is 0.35, set once on the development split and not moved since (§3.6).

Onset. A window cannot say where inside it a sound begins, and with the simple time-stamping just described, every onset on three hand-checked clips was late by +0.25 to +2.0s. Two standard pieces on the same model fix most of it. *Hysteresis*: once a sound reaches the bar, its start is moved back to the beginning of the unbroken stretch in which its score stayed above half the bar (an approaching helicopter scores 0.15–0.42 for 1.5s before crossing 0.35). *Prefix-silencing occlusion* inside the first window that fired: mute the first t seconds of that window for 26 values of t , score all 26 versions in one model call, and take as onset the smallest t at which muting removes a tenth of the model’s evidence (in logits) for that class. The measure is specific to the class, so speech in the same window does not shift it. Three alternatives were measured and rejected: finer windows and frame-level detectors (PANNs 10 ms, BEATs-strong 40 ms) miss about half the sounds; a class-activation map over BEATs’ 160 ms tokens moves every onset by exactly 0 or -1.5 s, because the model was fine-tuned only on the average over tokens, so no single token was ever trained to say where in time a sound is. Results in §6.

3.3 Stage 5: is the source on screen?

The proposal’s premise is that a picture is due only when the source of a sound is *not* on screen, so this question carries the system. It was first answered by a hand-written dictionary of about thirty labels mapped to CLIP prompts; a sound outside the dictionary could never be marked visible, so a laugh, footsteps or a telephone could never be silenced, whatever the frames showed. It is now asked at run time about the sound that was actually detected, from frames sampled across that sound’s own span.

Name it, do not confirm it. The VLM (Qwen2.5-VL-7B) is asked to name the thing in the frames making that sound, or “nothing”. A yes/no question to a VLM collects agreement rather than evidence; a name can be checked. What is in the frame is perceptual and is asked of the pixels; whether that thing makes this sound is world knowledge and is asked as plain text, so a loosely related object cannot count as the source without a table of which objects make which sounds.

The action, not the object. A baby in its mother’s arms was named as the source of the crying and silenced; a deaf viewer looking at it learns nothing about the sound. A second

question therefore asks whether the thing can be seen *making the sound now*. Three signals vote: open naming, the world-knowledge check, and an open description of the frames. Every two-way question is asked twice with the options swapped, and both answers must agree. This is needed because the 7-billion-parameter model favours a position: it answered “(b)” in 22 of 23 calls, whatever option (b) was. Visibility is judged per 5 s stretch, but one continuous sound gets one continuous picture even if its source enters the frame midway.

Speech as gate context. For each sound with speech within three seconds, one text-only forced choice asks whether the people on the soundtrack are reacting to it. A “yes” gives that sound priority when the panel is full and can admit a sound just below the bar. The transcript never enters a picture and never overrides the visibility decision. It is not asked for fewer than four transcribed words, since Whisper transcribes non-speech as phonetic junk. On the 100-clip run it was asked 77 times and answered yes twice.

The frames may choose between the detector’s own candidates. When two labels share both boundaries within 0.6 s and are not parent and child in the AudioSet label hierarchy (next subsection) – one sound with two names, from the same windows –, the VLM is shown the frames and asked which it is, with “cannot tell” keeping both. The audio proposes, the video picks, and nothing is added that the audio did not hear.

3.4 Stage 5: one picture per source

The detector emits label families (*Laughter*, *Snicker*, *Chuckle*) and a clip once produced three near-identical pictures of people laughing. Three general mechanisms were tried. Label embeddings fail because single words are too similar to each other (cosine similarity 0.90 for *Dog/Cat*, higher than 0.80 for *Laughter/Snicker*, so similarity does not track sameness). Asking the VLM whether two depictions show one thing fails because it answers the question literally: “different” for 90% of pairs, including an owl on a branch against an owl’s hoot. A similarity threshold fails because it must be re-tuned whenever a prompt elsewhere changes, which happened three times. What works is the detector’s own taxonomy: the AudioSet ontology (CC BY 4.0), stored in the repository as child-to-parent links, and the rule that one label must be an *ancestor* of the other – a shared parent is not enough, since *Animal* would swallow *Dog* with *Sheep*. The rule was correct on all 14 pairs the demos produced, with no threshold. A family label explains a member only where they overlap in time (interval subtraction), so an all-clip *Bird* does not swallow a five-second *Owl*. Depiction similarity survives as a secondary signal at 0.80, and only for pairs that overlap in time.

3.5 Stage 6: what the picture shows

Depiction. For each sound the VLM writes a two-to-five word event phrase (“glass shattering”, “audience clapping”), built from the detector’s label, the sound’s kind as seen in the frames and a *place* of at most four words asked for separately. A place replaced the one-sentence scene description after that sentence’s objects were glued onto every sound: an *Air horn* became “air horn blaring from silver van” and rendered as a picture of a van. The transcript never reaches the prompt: when it did, a *Hiccup* became “Hiccup on the phone, slow down, how many”. No example phrases are given; the prompt is built from the detector, the audio and the video only.

Validation by choice. A yes/no check approved “man on motorcycle drives past silver van” as a picture of *Laughter*. A comparative check (is the phrase closer to the label than the scene sentence was?) was right on all eight cases it was built on but only 11 of the next 17, because it was really rewarding shorter phrases. The working check asks the VLM to *choose*: given the phrase and the other sounds detected in the same clip plus “none of them”, which sound does it show, asked in both orderings.

Rendering and panel. FLUX.1-schnell renders the phrase at 768 px on a plain white background (chosen over PixArt- Σ on twelve real prompts). The panel’s rows depend on how

many sounds are heard at the same time, not on how many the clip contains: rows are the largest number of sounds heard at once (at most three), a picture is held at least 1.5s, bursts closer than 0.8s are one appearance, and a cell is empty when its sound is absent. The first design parked every sound’s picture beside the video for the whole clip, dimmed when silent; a panel that always looks the same carries no information, and a picture on screen while its sound is absent makes a claim a deaf viewer cannot check.

3.6 Design process

Before 2026-09-13, fixes were made on whichever clip was being inspected, and thresholds were re-tuned clip by clip. Since 2026-09-13 the rules are: thresholds are set once on the development split; every failure seen in a demo is logged in a failure catalogue with clip, stage and confidence; a design change needs three entries of one pattern or a metric move on the development split; demos illustrate and never tune; and every change is general and re-measured. The catalogue holds 18 entries, seven fixed as general bugs, one decided (the place veto, off after two right and two wrong in four uses), the remaining ten are open, each seen once; three sightings are needed before a change.

4 Benchmark

Clips and labels. 274 clips were hand-labelled into four scenarios: *unseen_ambient* (an ambient sound whose source is not on screen), *mixed* (some sources visible, some not), *seen_ambient* (every ambient sound’s source is on screen) and *no_ambient* (speech or music only). The central labelling rule is the redundancy test used by professional captioners (DCMP): not “is the physical source in frame?” but “does the picture already tell the viewer that this sound is happening?” – wind roar over a rider filmed on a moving motorcycle is *seen*, and so are waves when the sea fills the frame. The main runs use 100 clips, 25 per scenario, so that abstention is scored on as many clips as augmentation.

Yield. The proposal named 300 clips. Four independent sourcing strategies (keyword scraping, AudioSet temporally-strong segments, curated film scenes, UnAV-100 overlap moments) and a three-stage automatic filter all landed in the same 15–20% band of usable positives, and yield fell as filtering grew. Yet 36% of clips that contain ambient sound at all are unseen or mixed: the phenomenon is not rare, the acquisition funnel discards it. The first 209 tags predate any model-based filtering and are the unbiased sample.

Label stability. The annotator re-labelled sixty clips in shuffled order without seeing the earlier label. The two labels agreed on 78% (Cohen’s $\kappa = 0.60$; 1 is perfect agreement, 0 is chance). The instability is in the positive class (*unseen_ambient* reproduced 12/20 against *seen_ambient* 19/20), which is the class the gate is judged on.

Check for over-tuning on the test set. We split the labelled clips in half (equal scenario mix), tuned every threshold on one half, tested the frozen values on the other, and repeated with five random splits: the held-out half scored higher than the sweep half in four of five, and the sweep chose the same corner of the grid each time. There is no optimism to remove, because the thresholds are not doing the work; the gate is limited by what a vision–language model can answer, not by its configuration.

4.1 External check: DCASE 2025 Task 3 on/off-screen labels

visibility check vs. DCASE 2025 gold (258 events)	
agreement	66.7%
off-screen recall (sound correctly not silenced)	97%
on-screen recall (visible source correctly silenced)	35%

The check is timid in one direction: it almost never silences a sound whose source is off screen, and silences only a third of those whose source is on screen. The DCASE on/off-screen label is geometric – the source’s direction of arrival falls inside a 100-degree field of view cut from a 360-degree capture – not “the viewer can see it making the sound”, so part of the on-screen shortfall is the check answering a stricter question than the label.

5 Evaluation protocol

5.1 Three references

model-derived (v2, v3) as specified in the proposal: written by the system’s own VLM from the detector’s own events. Found circular on 2026-09-14 (review): it rewards whichever system repeats the detector.

human-grounded (v2_grounded, v3_grounded) the same reference corrected by the annotator’s label; only computable on labelled clips; by construction it raises a system that stays silent more than the baselines (see the rubric asymmetry, §6.4).

independent (v2_indep, v3_indep) built by models that never see the system’s output. (1) Qwen2-Audio, an audio language model, lists what it hears, on the raw audio and on a version with vocals removed by Demucs (a source-separation model), in 10-second chunks; the two lists are merged and speech items dropped. (2) CLAP, an audio-text matching model, keeps a claim only if this clip’s audio ranks in the top 20 of the 100 benchmark clips for that claim’s text, with positive similarity. (3) Idefics3-8B lists what is visible in the frames. (4) Phi-3.5-mini writes the reference from these lists (Llama-3.1-8B is licence-gated and was not accessible to the account). (5) A sentence-similarity check (MiniLM, threshold 0.50, uncalibrated) removes any sentence not supported by the verified list. Two details: the vocals-removed audio alone was empty on a third of the clips because Demucs pulls tonal sounds into the vocals stem, hence the merge; and the first form of step (2), a mean $+2\sigma$ bar over random text decoys, left 58 of 100 references empty against 25 clips tagged *no_ambient*, so both forms are reported. It was built to be the corrected primary; §6 shows it cannot score the gate decision, so the human-grounded reference remains the one that does.

5.2 Baselines

Two baselines are used. *Blind audio-to-image* draws a picture for every sound heard, with no gate. *Audio captioning* writes a text description instead of a picture. Neither looks at the video.

5.3 Statistics

For each comparison we report per-clip wins, ties and losses; a bootstrap 95% confidence interval (CI) on the mean difference (5000 resamples); and an *oracle-gate* bound: the score if the human label were used as the gate and the best system’s picture shown whenever one is due (`scripts/paired_stats.py`).

6 Results

6.1 Main comparison

run	reference	proposed	blind a2i
v2	model-derived	2.27	2.97
v2_grounded	human-grounded	2.90	3.14
v2_indep_list	independent (list-and-match visibility; variant explained below)	2.41	2.96
v3	model-derived	2.64	3.04
v3_grounded	human-grounded	3.17	3.16
v3_indep	independent	not run: the reference ca	

6.2 Paired statistics

run	proposed vs. blind: win / tie / loss	mean diff	95% CI	oracle gate
v2	11 / 52 / 37	−0.70	[−0.98, −0.43]	2.89
v2_grounded	22 / 54 / 24	−0.24	[−0.47, −0.02]	3.19
v2_indep_list	12 / 59 / 29	−0.55	[−0.82, −0.30]	2.88
v3	12 / 61 / 27	−0.40	[−0.66, −0.16]	3.01
v3_grounded	21 / 64 / 15	+0.01	[−0.15, +0.18]	3.25

On v2 the blind baseline scores higher than the gated system under every reference, with the interval excluding zero each time; the circular reference was not what held the gate back. On v3 – the shipping configuration: BEATs, per-sound visibility, event depictions, ontology deduplication – the gap closes to a tie under the human-grounded reference (+0.01, 21 wins to 15 losses) while remaining −0.40 under the model-derived one. The oracle row is the ceiling any gate could reach on this benchmark with the best system’s output where a picture is due: v3 sits 0.08 below it. The gated system also beats the captioning baseline on v3_grounded (+0.28, [+0.10, +0.46]).

The independent reference is not neutral either. Its silence decision (“nothing beyond the picture”) agrees with the human tag on 55 of 100 clips – chance. The list-and-match visibility step (Idefics3 asked for a list of sound-capable things, one word back per clip; MiniLM cosine to each verified sound) calls almost every sound unseen, so on seen-ambient clips the reference says the viewer would miss the sound and the blind baseline is rewarded for showing it (seen_ambient: blind 3.08, proposed 2.48). A per-sound question to the same VLM (“is the thing producing *X* visible? answer visible: <name> or not visible”) was tried next and scored the same 55%, because the rule “silent only if every verified claim is visible” fails whenever one of five to eight audio-LM claims (*video game sound*, *wind noise (microphone)*, *clock ticking*) is not; that variant was not judged (the judge is run only on a reference whose silence decisions agree with the human label on at least 70% of clips). The reference’s remaining weakness is therefore the same question the system itself finds hardest – whether a sound’s source is on screen – and the honest reading is that no automatic reference built here can score the gate decision; the human-grounded reference remains the only one that can, with the asymmetry already stated.

6.3 By scenario group

The four scenarios were defined in the proposal before any system was run; the pooled score is reported first and then split by whether a picture is due (unseen and mixed clips) or not (seen and no-ambient clips), with the same judge score in both groups.

run	group ($n = 50$)	proposed	blind	caption	proposed – blind
v2_grounded	picture needed	2.26	2.92	2.66	−0.66 [−1.06, −0.28]
	no picture needed	3.54	3.36	2.92	+0.18 [+0.00, +0.34]
v3_grounded	picture needed	2.78	3.02	2.76	−0.24 [−0.54, +0.04]
	no picture needed	3.56	3.30	3.02	+0.26 [+0.12, +0.40]

The split makes the finding precise: the gate does the right thing when nothing is missing and still fails when something is. Of the 15 clips where the blind baseline beats v3, most are a picture withheld – the visibility check named a source that was on screen but not visibly making the sound, or the sound did not reach the display bar – and a withheld picture scores 0 where a redundant one scores about 3. Under that rubric the gate must be right about four times in five when it stays silent merely to break even, which is why a perfect gate only narrowly beats the blind baseline (3.25 vs 3.16). §6.7 reports the one change this licenses: setting the silence rule on the development split with the rubric’s own costs.

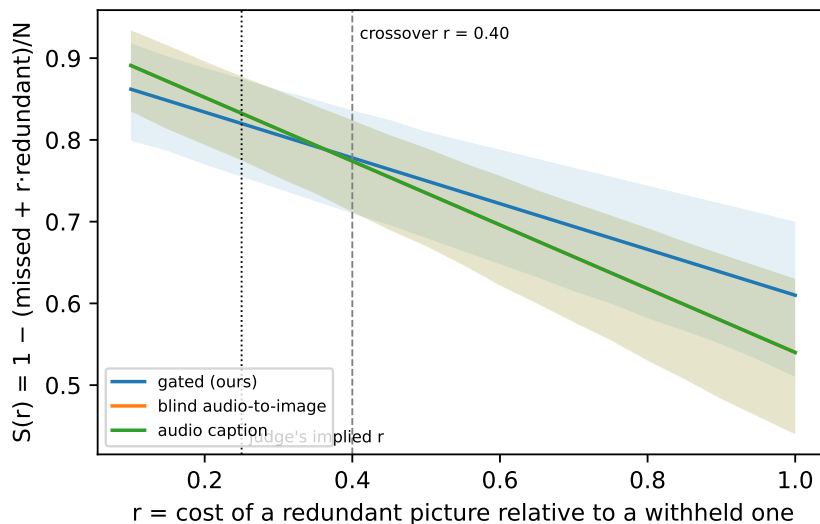
6.4 Cost sensitivity of the gate decision (secondary, post-hoc)

The pre-registered protocol prices the two errors asymmetrically: a withheld picture scores 0 where one would have scored up to 4, a redundant picture about 3 where silence scores 4 – a redundant picture costs roughly a quarter of a withheld one. That ratio is a property of the rubric, not of deaf viewers; captioning practice omits sounds whose source is visible because redundancy has a real cost (§2). We therefore add, as a secondary analysis declared after the main result, a judge-free score of the clip-level decision with the ratio as an explicit parameter. Per clip and system, *showed* means the system produced at least one augmentation and *due* means the human tag is unseen or mixed;

$$S(r) = 1 - \frac{\text{missed} + r \cdot \text{redundant}}{N},$$

with r the cost of a redundant picture relative to a withheld one, swept from 0.1 to 1.

system	showed	hit	missed	redundant	precision	recall	$S(0.25)$	$S(1)$
gated (ours)	67	39	11	28	0.58	0.78	0.820	0.610
blind audio-to-image	82	43	7	39	0.52	0.86	0.833	0.540
audio caption	82	43	7	39	0.52	0.86	0.833	0.540



At the judge’s implied ratio ($r \approx 0.25$) the two systems tie (-0.012 , 95% CI $[-0.057, +0.025]$); the gated system is preferred from $r = 0.40$ upward and leads by $+0.070$ at $r = 1$ (CI $[0.000, +0.150]$). The claim this licenses is conditional: *the gate is preferred whenever a redundant picture costs more than about forty percent of a withheld one*; whether real viewers’ costs exceed that is an empirical question this thesis does not answer. The reviewers’ accessibility reading – redundancy erodes trust in the channel and compounds over a session, so its cost is likely above the rubric’s quarter but below parity, and higher for alarm-like sounds than for ambience – is an argument, not a measurement.

This metric scores a clip as correct if the system showed at least one picture when the clip contains an off-screen sound, or none when it does not. It does not assess whether the picture was shown at the right moment, for the right sound, or with the right content; the pre-registered judge score covers those. It also exposes a blind spot of the judge: on the seven clips where the detector heard nothing on a clip tagged unseen or mixed, the grounded reference itself reads “nothing beyond the picture” and every system scores 4, so a detector miss is invisible to the judge; here it counts as a miss for all three systems alike. The caption baseline coincides with the blind one at the decision level (same detector, no gate).

6.5 Onset timing

Against seven hand-labelled onsets on three clips the plain stamp was late by $+0.9$ s on average and the shipping method by $+0.3$ s; that is an anecdote, so the same question was put to DCASE 2025 Task 3, whose events carry a start frame at 100 ms. From 255 gold events (nine non-speech classes, onset inside the clip) the same BEATs scores were converted to onset times by three methods; an event counts as matched when a detection in the same AudioSet family starts within 1.5 s of it. Every error below is measured on the matched events only – 66 of 255 (26%) at the display bar – so the table describes where a picture is stamped when a sound is detected, not how often it is.

stamping	matched	mean signed	MAE	≤ 0.25 s	≤ 0.5 s
sliding-window stamp (before 09-14)	50	$+0.29$	0.43	44%	68%
+ hysteresis	66	$+0.22$	0.35	58%	74%
+ hysteresis + occlusion (shipping)	66	$+0.16$	0.35	45%	80%

Hysteresis is the gain: on the 50 events both methods match, the mean absolute error (MAE) falls from 0.43 to 0.31 s, and it matches sixteen more events, because a sound that reaches the bar late is now traced back to where it began. Occlusion adds nothing on average: on the 65 events both match it is better on 25 and worse on 39 with an identical MAE of 0.34 s (paired bootstrap difference 0.00, 95% CI $[-0.11, +0.10]$). It trades a smaller late bias and more onsets within half a second for fewer within a quarter-second; on knocks, where hysteresis alone is within 0.08 s, it moves the onset $+0.26$ s late. The seven-clip number therefore overstated the method, and the honest statement is that the shipping onset is within half a second for four detected events in five. Detector recall on these five-second clips is the larger limit: laughter, footsteps and domestic sounds rarely reach 0.35 on BEATs. Whether to keep occlusion is a development-split decision left open (§3.6); it does not alter detection or gating, only where a picture is stamped.

6.6 Ablations recorded during development

Retrieval against generation (earlier configuration). On the same 100 clips with the first pipeline, retrieving a stock photograph for the label beat SDXL by $+0.17$ (gated) and $+0.18$ (blind), while the captioning baseline, which uses no images, was unchanged to two decimal

places, as expected; 55% of the retrieved images carried a non-commercial or no-derivatives licence and 595 uses drew on 60 distinct images, which is why generation was kept.

Judge reliability. A second judge model from a different family agrees with the primary one at $\kappa = 0.753$ (97% within one point) and reproduces the gated-versus-blind gap; the captioning baseline moves by +0.61 under it, so any ranking involving the text baseline is a property of the judge.

Detector. Eleven of fourteen complaints from a review of eight demo clips traced to PANNs’ labels; BEATs resolved every one on the same clips, and the display bar moved with it.

Negative results kept. A class-activation map over BEATs’ tokens moves every onset by exactly 0 or -1.5 s and cannot work; a plausibility veto on the place (“a dog in a parking lot”) was right twice and wrong twice in four uses and is off; a comparative depiction check that scored 8/8 on the cases it was built on scored 11/17 on the next nine, because it measured phrase length.

6.7 Where the gate’s errors come from

The loss on “picture needed” clips licenses one change under the process rules: set the gate’s silence rule once on the development split, scoring each setting with the rubric’s own costs (a withheld picture scores 0 where one would score up to 4, a redundant one about 3 where silence scores 4: -4 and -1 per clip). The three visibility votes of every sound at or above 0.25 were cached for the 156 development clips with a video file, and sixteen settings were scored from the cache: silence by majority or by unanimous vote, display bar 0.25–0.40, the kinds-of-a-visible-source rule (a sound is also silenced when a visible object is a kind of thing that makes it) on or off.

rule	bar	kinds	accuracy	cost/clip	missed / redundant
majority	0.25	on	32.1%	-0.679	0 / 106
majority	0.30	on	38.5%	-0.615	0 / 96
majority	0.35	on	44.9%	-0.571	1 / 85
majority	0.40	on	48.1%	-0.577	3 / 78
unanimous	0.35	on	39.1%	-0.609	0 / 95
majority	0.35	off	43.6%	-0.583	1 / 87

The shipping setting is the optimum on cost, and every setting that reduces redundancy adds misses. Two things follow. First, on the development split the gate is not timid: it misses one of 23 unseen clips and shows a redundant picture on 85 of 133 seen or no-ambient clips. Second, the sweep could not observe the failure it was launched to fix – all 25 mixed clips are in the test set, none in development – so no tuned variant was run; the shipping numbers stand.

Reading the errors instead. The cached votes explain both directions without a GPU. Of the eleven test clips where the gated system showed nothing and a picture was due, seven are the detector: nothing reached the bar, because the ambient sound sat under speech or music (crowd hubbub 0.22 under *Speech* 0.84; rain 0.23 under *Music* 0.81; drums 0.29 under *Music* 0.65) or was a texture at 0.24–0.31. The other four are *Vehicle* in street scenes, where cars are in frame and every stretch was unanimously judged visible with a stable name (“white car”, “taxi on road”): under the annotation’s own redundancy rule the gate’s silence is defensible and the tag is the debatable one. None is a timing mismatch. On the development side, of the hundred redundant pictures the VLM could not silence because it saw no source, about eighty are confident detector *phantoms* (sounds reported with a high score that are not in the audio at all) with no source anywhere – a whale at a Christmas market (0.36), roaring cats at a helicopter arrival (0.51), an electric toothbrush in an alarm clip (0.73), glass at 0.87 in a bombardment – and the VLM is right that nothing visible makes them. Only about twenty are real sources the

VLM said were absent (thunder in a storm, an alarm in a car-alarm clip), and those expose a definitional gap between the annotation’s redundancy test (“does the picture already tell the viewer this is happening?”) and the question the VLM is asked (“name the thing making the sound”).

The bottleneck is therefore the detector on both sides – recall masked by speech and music, and confident false positives – more than the visibility judgment. A second opinion from an independent audio model is the model-based answer to the phantoms: CLAP scores the same two seconds against all 527 AudioSet names collapsed to ontology families, and a detection survives only if its family is among CLAP’s top k , with k set once on DCASE gold as the smallest value retaining 95% of gold recall. The check was calibrated on the 255 DCASE gold events by ranking each event’s family among the 328 ontology families for the event’s own burst; the smallest k reaching 95% recall had to be at most 30 for the check to reject anything, both declared before the run. Recall was 10% at $k = 1$, 32% at $k = 10$ and 50% at $k = 30$, with per-class median ranks from 3 (water tap) and 16 (domestic sounds) to 53 (door), 70 (laughter) and 261 (footsteps): no usable k exists. Together with the same model verifying claims at chance as an evaluation reference (§5), this is reported as a measured limitation of zero-shot audio–text matching on short ambient events in polyphonic audio, not as a tuning failure; the shipped system uses BEATs alone.

The remaining candidate, source separation in front of the detector (Demucs; re-detect on the residual, keep the higher score), was assessed and not run: the residual removes speech only, so four of the seven masked misses, which sit under music, are untouched, and taking the higher of two scores can only add detections, so it raises the phantom rate by construction. It is the first future-work item, with its acceptance bars stated in §7. The finding that stands is that in polyphonic video a clip-level tagger’s confidence is not evidence that a sound is in the scene; the detector, not the visibility judgment, bounds the headline result, and no threshold inside either recovers it.

7 Discussion

Which audio information contributes most when shown visually? Discrete, sourced events – sirens, alarms, bells, a baby crying, glass breaking – are what the judge rewards and what a picture can carry; continuous textures (wind, traffic, rain) are where the detector, the visibility check and the reference all disagree most, and where the blind baseline’s redundant picture costs it least. The benchmark’s yield says the same thing from the other side: only 15–20% of candidate clips contain an ambient sound whose source is off screen, but a third of clips with any ambient sound do.

What granularity? The detector’s own taxonomy. Gating on the AudioSet family and depicting the specific member – a fire-engine siren, not a siren – was the one deduplication rule that generalised (14/14 with no threshold), after label embeddings, a VLM judgement and a similarity threshold each failed for a stated reason.

Can generated augmentations improve accessibility beyond subtitles? Not shown here; the evaluation is a proxy. What is shown is that under a rubric that scores what a viewer would learn, the gated system ties a baseline that draws everything and beats the text baseline, that it wins where nothing is missing and loses where something is, and that the loss is mostly the detector’s – sounds masked by speech or music that never reach the bar – with the visibility judgment second (§6.7).

How should such systems be evaluated? With a reference the system cannot have written. The proposal’s reference is built from the detector’s events by the system’s own VLM and favours the system that repeats the detector by half a point. A reference built by independent models removes that bias but introduces its own: its on-screen decision is at chance, so it rewards showing. The only reference that scores the gate is the one corrected by the human label, and it

is not available to a deployed system. The rubric’s asymmetry must be stated whenever a gated system is compared with a blind one.

How does the pipeline compare with audio-to-visual generation? It ties it on the pooled score and beats it by +0.26 where no picture is due, at the cost of −0.24 where one is. A perfect gate would gain 0.09 over the blind baseline on this benchmark. The gate’s value is therefore real but small under this rubric, and it lies entirely in what it does not show.

8 Limitations

Stated here rather than left for a reader to find; `LIMITATIONS.md` in the repository keeps the full list in step with the code.

One annotator ($\kappa = 0.60$ against their own earlier labels, unstable on the positive class). **One judge decides the headline**; a second reproduces the gated-versus-blind gap but not the text baseline’s rank. **100 test clips, 25 per scenario**: read the paired intervals, not the means. **No human study**: a judge scoring a description of a picture is a proxy for comprehension. **The references are not neutral**: the model-derived one is circular, the independent one decides visibility at chance, the grounded one is computable only with labels. **The detector’s vocabulary and recall**: AudioSet has no “phone alert”; a siren under a film score is heard only at its close-up; on five-second DCASE clips only 26% of gold events reach the display bar. **The visibility check is timid toward on-screen sources** (35% on DCASE gold) and unstable to a half-second shift in frame sampling. **The onset method was over-claimed** on seven hand-labelled onsets and is reported on 255. **The evaluation software produced plausible wrong numbers** repeatedly – a charitable score for an empty augmentation, a similarity that returned 0.0 for everything, an audio model that never received its audio – each caught by reading logs, each now guarded.

9 Conclusion and future work

A deaf viewer of a video misses the sounds whose sources are not on screen, and no existing channel supplies them automatically. This project built the pipeline that does – detection, a visibility decision from the frames, an event depiction, a picture beside the video – from off-the-shelf models with no training, a benchmark of 274 labelled clips in the proposal’s four scenarios, and an evaluation protocol whose reference had to be rebuilt twice before it could be trusted. The shipping system ties a baseline that draws every sound it hears and beats the text baseline; it wins where nothing is missing and loses where something is, because the rubric prices a withheld picture at four times a redundant one and the visibility question is the hardest a vision–language model is asked here.

Next, in order:

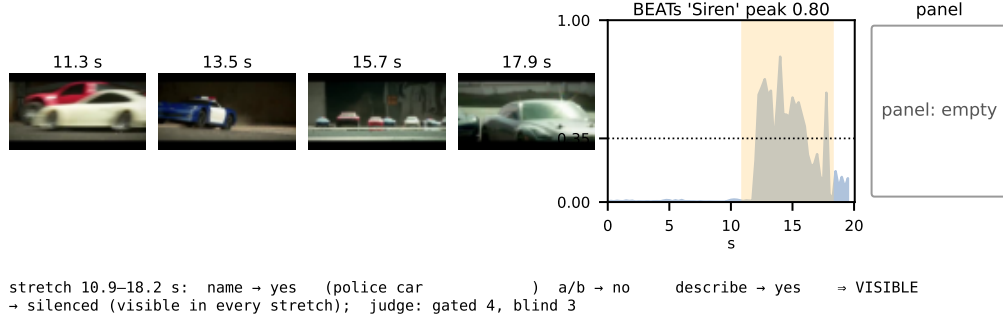
1. a detector-only benchmark on the labelled clips: recall of off-screen events against the phantom rate, independent of gate and judge;
2. against it, a speech-aware detector – BEATs on a Demucs residual with the higher of the two scores – accepted only if DCASE recall on speech-overlapped events rises by ten points, phantoms per class rise by no more than two, and no threshold is re-tuned; or a polyphonic detector trained on mixtures;
3. an open-vocabulary audio model for the sounds the 527 classes have no word for;
4. a second annotator;
5. a study with deaf viewers, without which the accessibility claim rests on a proxy.

References

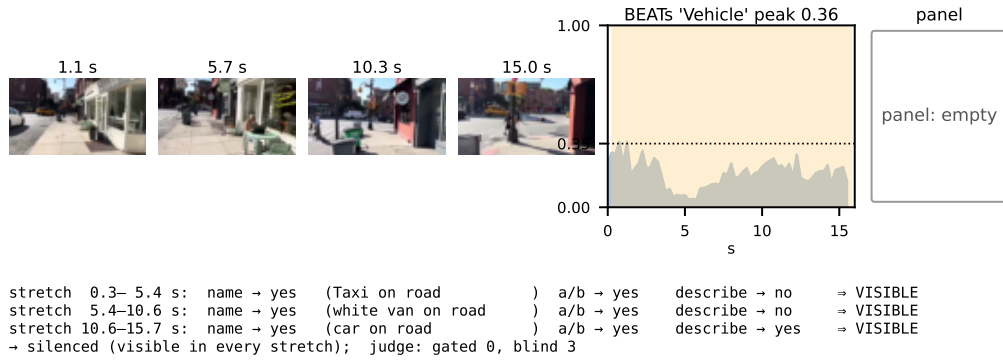
- [1] J. F. Gemmeke et al., “AudioSet: An ontology and human-labeled dataset for audio events,” *ICASSP*, 2017.
- [2] Q. Kong et al., “PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition,” *IEEE/ACM TASLP*, 2020. arXiv:1912.10211.
- [3] Y. Gong, Y.-A. Chung, J. Glass, “AST: Audio Spectrogram Transformer,” *Interspeech*, 2021. arXiv:2104.01778.
- [4] S. Chen et al., “BEATs: Audio Pre-Training with Acoustic Tokenizers,” *ICML*, 2023. arXiv:2212.09058.
- [5] C. D. Kim et al., “AudioCaps: Generating Captions for Audios in The Wild,” *NAACL*, 2019.
- [6] K. Drossos et al., “Clotho: An Audio Captioning Dataset,” *ICASSP*, 2020.
- [7] B. Elizalde et al., “CLAP: Learning Audio Concepts from Natural Language Supervision,” *ICASSP*, 2023.
- [8] W. Chen et al., “SLAM-AAC: Enhancing Audio Captioning with Paraphrasing Augmentation and CLAP-Refine through LLMs,” 2024. arXiv:2410.09503.
- [9] Y. Chu et al., “Qwen-Audio / Qwen2-Audio: Advancing Universal Audio Understanding,” 2023–2024. arXiv:2311.07919.
- [10] C. Tang et al., “SALMONN: Towards Generic Hearing Abilities for Large Language Models,” *ICLR*, 2024.
- [11] S. Sakshi et al., “MMAU: A Massive Multi-Task Audio Understanding and Reasoning Benchmark,” 2024. arXiv:2410.19168.
- [12] Qwen Team, “Qwen2.5-VL Technical Report,” 2025. arXiv:2502.13923.
- [13] H. Liu et al., “Visual Instruction Tuning (LLaVA),” *NeurIPS*, 2023.
- [14] K. Sung-Bin et al., “Sound to Visual Scene Generation by Audio-to-Visual Latent Alignment (Sound2Scene),” *CVPR*, 2023. arXiv:2306.11504.
- [15] G. Yariv, I. Gat, S. Benaim, L. Wolf, I. Schwartz, Y. Adi, “Diverse and Aligned Audio-to-Video Generation via Text-to-Video Model Adaptation (TempoTokens),” *AAAI*, 2024. arXiv:2309.16429. <https://arxiv.org/abs/2309.16429>, code: <https://pages.cs.huji.ac.il/adiyoss-lab/TempoTokens>
- [16] B. C. Biner, F. M. Sofian, U. B. Karakaş, D. Ceylan, E. Erdem, A. Erdem, “SonicDiffusion: Audio-Driven Image Generation and Editing with Pretrained Diffusion Models,” 2024. arXiv:2405.00878.
- [17] D. Podell et al., “SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis,” 2023. arXiv:2307.01952.
- [18] Black Forest Labs, “FLUX.1,” 2024. <https://blackforestlabs.ai/>.
- [19] P. Revuelta, T. Ortiz, M. J. Lucía, B. Ruiz, J. M. Sánchez-Pena, “Limitations of Standard Accessible Captioning of Sounds and Music for Deaf and Hard of Hearing People: An EEG Study,” *Frontiers in Integrative Neuroscience*, vol. 14, 2020.

- [20] “Visualization of Speech Prosody and Emotion in Captions: Accessibility for Deaf and Hard-of-Hearing Users,” *CHI*, 2023.
- [21] D. Jain et al., “SoundWatch: Smartwatch-based Deep Learning Approaches to Support Sound Awareness for DHH Users,” *ASSETS*, 2020.
- [22] D. Jain et al., “HomeSound: An Iterative Field Deployment of an In-Home Sound Awareness System for DHH Users,” *CHI*, 2020.
- [23] D. Jain et al., “ProtoSound: A Personalized and Scalable Sound Recognition System for DHH Users,” *CHI*, 2022. arXiv:2202.11134. <https://arxiv.org/abs/2202.11134>
- [24] Described and Captioned Media Program (DCMP), “Captioning Key — Guidelines and Preferred Techniques” (Sound Effects, Music, Non-Speech Information sections). <https://dcmp.org/learn/captioningkey>
- [25] M. Arroyo Chavez, M. Feanny, M. Seita, B. Thompson, K. Delk, S. Officer, A. Glasser, R. Kushalnagar, C. Vogler, “How Users Experience Closed Captions on Live Television: Quality Metrics Remain a Challenge,” *CHI*, 2024. arXiv:2404.10153.
- [26] R. S. Kushalnagar, W. S. Lasecki, J. P. Bigham, “Captions versus Transcripts for Online Video Content,” *W4A*, 2013.
- [27] Y. Nishi, Y. Nakamura, S. Fukushima, Y. Arakawa, “LightSub: Unobtrusive Subtitles with Reduced Information and Decreased Eye Movement,” *Multimodal Technologies and Interaction*, vol. 8, no. 6, 2024.
- [28] A. Senocak et al., “Sound Source Localization is All about Cross-Modal Alignment,” *ICCV*, 2023. arXiv:2309.10724. <https://arxiv.org/abs/2309.10724>
- [29] S. J. Um, D. Kim, S. Lee, J. U. Kim, “Object-aware Sound Source Localization via Audio-Visual Scene Understanding,” *CVPR*, 2025. arXiv:2506.18557.
- [30] M. Yoshida, R. Togo, T. Ogawa, M. Haseyama, “Off-Screen Sound Separation Based on Audio-visual Pre-training Using Binaural Audio,” *Sensors*, vol. 23, no. 9, 2023.
- [31] H. Chen et al., “VGGSound: A Large-scale Audio-Visual Dataset,” *ICASSP*, 2020. arXiv:2004.14368.
- [32] T. Geng et al., “Dense-Localizing Audio-Visual Events in Untrimmed Videos (UnAV-100),” *CVPR*, 2023. arXiv:2303.12930.
- [33] Q. Huang et al., “MovieNet: A Holistic Dataset for Movie Understanding,” *ECCV*, 2020. arXiv:2007.10937.
- [34] A. Radford et al., “Learning Transferable Visual Models From Natural Language Supervision (CLIP),” *ICML*, 2021.
- [35] A. Radford et al., “Robust Speech Recognition via Large-Scale Weak Supervision (Whisper),” 2022. arXiv:2212.04356.

(a) correct silence: the siren's police car is on screen



(b) wrong silence: cars visible, the off-screen traffic still needed a picture



(c) correct picture: siren heard, no ambulance in frame

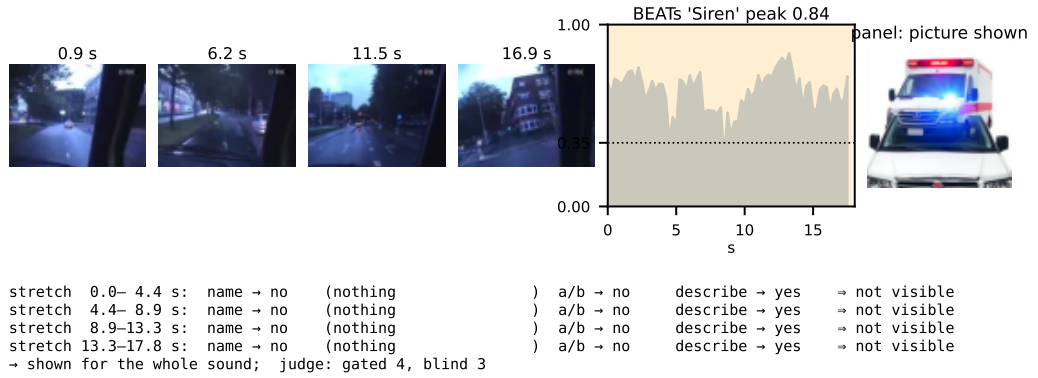


Figure 2: Three test clips end to end: four frames across the sound, BEATS’ confidence for the sound over time (shaded: the detected span; dotted: the 0.35 bar), the three visibility votes per stretch, and the panel. (a) The police car is on screen: the siren is silenced and the gated system scores 4 where the blind baseline’s redundant picture scores 3. (b) Cars are visible while the annotated sound is off-screen traffic: the gate silences *Vehicle* and scores 0 – a wrong silence costs four points. (c) No ambulance in frame: in every stretch the name and forced-choice votes say “nothing” (one vote of three), the picture “ambulance siren blaring” is shown for the whole sound.